

Set Up a Reverse Proxy in Apache

Read on to learn how to set up and configure a reverse proxy.



The Apache HTTP Web server is extremely popular, and is used by most enterprises and start-ups for their front facing servers. More commonly known as just Apache, it was developed by the Apache Software Foundation and was released in 1995. The most recent and stable version is 2.4.7, which was released in November last year. Apache is open source software and is available for most of the major operating systems, including UNIX, Linux, OS X, Microsoft Windows, etc. Let's now explore how to install Apache HTTP server and configure it to set up a reverse proxy on it.

Before proceeding to the actual set-up and configuration, I would like to explain what a reverse proxy is. According to Wikipedia, "A reverse proxy is a type of proxy server that retrieves resources on behalf of a client from one or more (internal) servers." In reality, the actual resources might be getting fetched from different internal servers, but the client will not be aware of it. For the client, it will appear as though those resources were generated from the server itself. To understand the scenario, consider the example shown in Figure 1. Generally, this job is done by a dedicated proxy server. But it may not be possible to have a dedicated proxy server always. Hence, we can use the Apache HTTP server to act as a proxy server.

Reverse proxies come with a lot of benefits. As mentioned on Wikipedia, the following are the major advantages of using reverse proxies:

1. Reverse proxies can hide the existence and characteristics of an origin server or servers.
2. Application firewall features can protect against common Web-based attacks. Without a reverse proxy, removing malware or initiating takedowns, for example, can become difficult.
3. In the case of secure websites, a Web server may not perform SSL encryption itself, but instead offload the task to a reverse proxy that may be equipped with SSL acceleration hardware. (See SSL termination proxy.)
4. A reverse proxy can distribute the load from incoming requests to several servers, with each server serving its own application area. In the case of reverse proxying in the neighbourhood of Web servers, the reverse proxy may have to rewrite the URL in each incoming request in order to match the relevant internal location of the requested resource.
5. A reverse proxy can reduce the load on its origin servers by caching static content, as well as dynamic content—also known as Web acceleration. Proxy caches of this sort can often satisfy a considerable number of website requests, greatly reducing the load on the origin server(s).
6. A reverse proxy can optimise content by compressing it in order to speed up loading times.
7. In a technique known as 'spoon feeding', a dynamically generated page can be produced all at one time and served to the reverse-proxy, which